Music Practice Robot | System Architecture

Desktop robot / Dual microphones / Touchscreen / Single-servo head / GPT-powered coaching

01 CLOUD COACHING LOOP

Recording Upload
Recorded files / streaming audio
Target score + performance context

GPT Analysis
Audio + pitch / rhythm analysis
Performance review + practice advice

Text-to-Speech (TTS)
Convert coaching advice to speech
Prepare audio for robot playback

Feedback to Robot
Coaching text + spoken feedback
On-screen display + audio playback

Coaching text + audio download

Recording upload

Wireless Coprocessor
e.g. ESP32-C6 | Wi-Fi 6 / BLE 5
Upload / receive feedback / update library

02 SENSING & STORAGE

Dual Microphones
Dual MEMS array
Synchronized capture

ES7210
Multichannel audio ADC
Synchronized sampling

I2S / I2C

SDIO / UART*

ESP32-P4 Main MCU
Dual-core: audio + UI / networking
Pitch / sound direction / score matching
Software AEC / noise reduction / AGC
Recording buffer / upload / feedback
UI rendering / audio / motion control

32 MB PSRAM + 32 MB Flash
Audio / frame buffers; firmware / score library

Memory bus

Micro SD Card
Long-session recording / music library expansion

SDMMC

03 DISPLAY, AUDIO & MOTION

DSI / I2C

4.3-inch IPS Capacitive Touchscreen
480 × 800 | Score / performance / coaching
MIPI-DSI display + I2C touch

I2S / I2C

ES8311 → Amplifier → Speaker
I2S audio output + I2C configuration
Spoken advice / voice replies / example audio

PWM

5 V Servo → Head Rotation
20 kg·cm class | Face sound source / gestures

04 POWER SYSTEM (VERIFY AGAINST SELECTED HARDWARE)

3.7 V Li-ion Cell
MX1.25 connector
Size for required battery life

Power Management
Charging / battery protection
Verify operation while charging

Regulated System Rails
MCU / wireless / display / audio
Set rail voltages per device requirements

5 V Boost & High-Current Branch
Dedicated servo branch + bulk capacitance
Amplifier rail depends on selected part

LEGEND Solid: data / control. Dashed: power. Use a dedicated servo supply branch with a common MCU signal ground.

NOTES AEC requires a playback reference. Memory sizes follow the source design. Verify audio support for the selected GPT model.

SCOPE Camera removed. Recording upload, cloud analysis, returned coaching and spoken playback included. *Board-dependent interface.